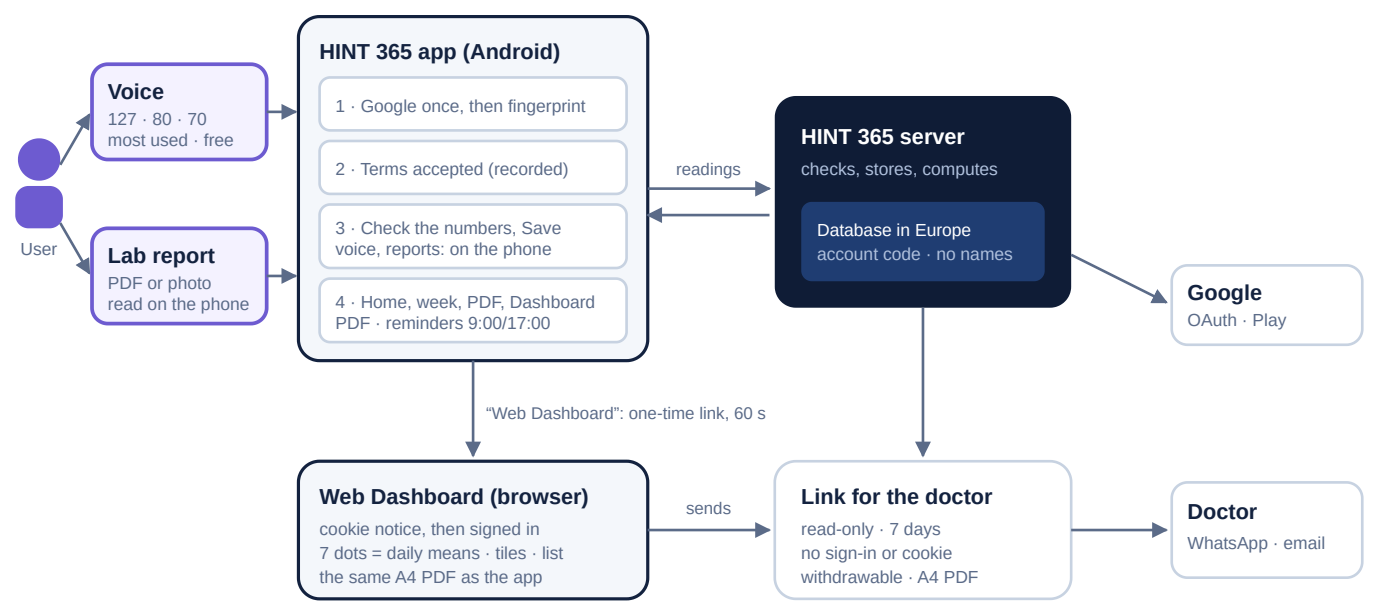


Functional overview

What the app does, who does what and in which order. A blood-pressure diary for Android, with a web dashboard and reports for the doctor.

The flow from the phone to the doctor



The user's path

1. **Installs** the app from the public link `/download` (later from Google Play).
2. **Signs in with Google** once, through OAuth 2.0 / OpenID Connect (a signed ID token, never the password), and gives privacy consent.
3. **Accepts the terms of use**, binding, at the first installation and after every update: without acceptance the app does not open. Every acceptance is recorded (pseudonymous account code, phone, versions, text fingerprint, date).
4. **Every day** unlocks with the phone's fingerprint or PIN, and stays signed in until "Sign out".
5. **Imports lab reports** by itself: PDF or photo read on the phone, every test (also outside the catalog), one report date; any doubt: nothing saved; never twice.
6. **Records** a reading: by voice or with a photo of the monitor (Scan with AI: 15-day trial, then 4.99 \$ a year; **SYS**, **DIA**, **PUL** checked and confirmed on screen). The photo is never stored (terms v21).
7. **Looks back**: last reading, the week; "Go to your full reports" at the end of Blood pressure or Lab results opens that page of the Web Dashboard in the browser (Blood pressure first: chart, eight tiles, morning/evening balance, pulse; then Lab results).
8. **Shares**: the PDF downloaded on the phone, or Web Dashboard → "Send to the doctor".
9. **Changes phone**: signs in with Google and finds everything. **Signs out or deletes the account** at any time.

Costs and who is paid

What	Cost	Paid to
HINT 365 app, Web Dashboard, readings	Free	Nobody
AI features (Scan)	15 days free, then 4 \$ a year	The app's owner, through Google Play

The app is free; only the AI features are paid (2.50 \$ of AI a year per subscriber).

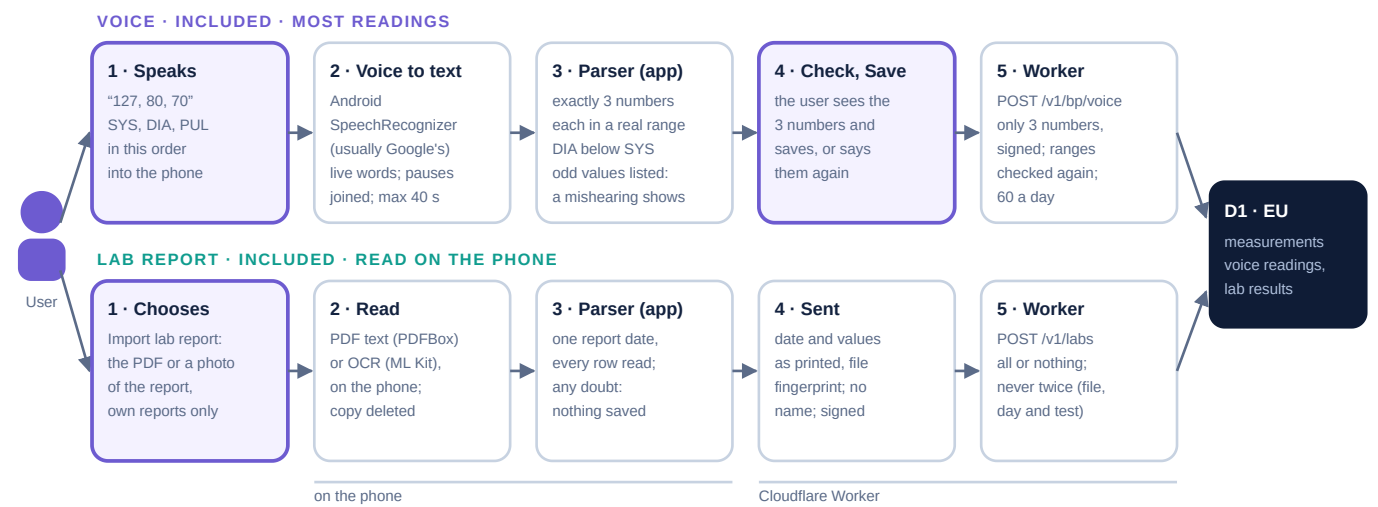
Fixed rules

- No diagnosis, no advice, no judgement of the values: no coloured zones, no "normal".
- Labels always **SYS**, **DIA**, **PUL**. Pressure and pulse never in the same chart.
- Charts: 7 days, one dot per day = the mean of that day's readings (explained before the charts), value next to it, axis with day numbers only. Tables show AM/PM.
- The same A4 PDF from the app and from the web.
- No red anywhere: **SYS** violet, **DIA** teal, **PUL** amber. No colour means "good" or "bad".
- Scan with AI (terms v21): 15-day trial, then 4.99 \$ a year (2.50 \$ of AI a year); the owner's key only, photo never stored.
- Owner's powers only from the owner's phone (`owner_key` ; new phone: `OWNER_CODE`).
- Readings kept at most 365 days, deleted nightly; acceptances stay as proof.
- The app stays simple: the owner's controls are only in the web Admin area.

Input: voice and lab reports

How data enters HINT 365: blood pressure by voice or by a photo of the monitor (Scan), lab results from a report. Voice and lab reports are read on the phone; the Scan is read with AI (terms v21): through the Worker to Anthropic with the owner's key.

From the user to the database the input always starts on the user's side



Purple frame: a step done by the user. Voice: the numbers are recognised on the phone and only the three numbers travel. Lab report: the document stays on the phone; only the date and the values as printed travel.

The differences

	Voice (blood pressure)	Lab report
Who can use it	everyone, included in the app	everyone, included in the app; own reports only (terms v18)
What the user does	says the three numbers	picks the laboratory's PDF or a photo of the report
Where it is recognised	on the phone: Android speech recognition (usually Google's service), then HINT 365's own parser (<code>parseSpoken</code>)	on the phone: PDF text (PDFBox) or on-device OCR (ML Kit), then <code>LabsParser.kt</code>
What leaves the phone	only the three numbers (the sound goes only to the phone's speech service)	the report date and the results as printed, and a keyed file fingerprint; never the document or a name
Cost	none	none
Checks	exactly 3 numbers; SYS 50–260, DIA 30–160, PUL 30–220; DIA below SYS; odd values listed; user confirms; ranges again on the server	one report date and every row understood, or nothing saved; server: schema, same file or same test and day never twice
Typical errors	a misheard number, noise, numbers said in the wrong order	an unclear photo, two dates on the document, an encrypted PDF
Limits	60 a day	PDF up to 15 pages, 20 MB
Stored as	<code>measurements</code> , <code>source voice</code> or <code>photo (Scan)</code>	<code>measurements</code> , <code>kind lab</code> ; <code>lab_files</code> fingerprint
Code	<code>Listen.kt</code> , <code>Core.kt</code> (<code>parseSpoken</code>) , <code>POST /v1/bp/voice</code>	<code>Labs.kt</code> , <code>LabsParser.kt</code> , <code>POST /v1/labs</code>

Scan (photo of the monitor): camera → `POST /v1/bp/photo/read` (trial or subscription, else `402 scan_locked` ; 3 a day, and the 2.50 \$ yearly allowance in `scan_usage` , else `429 scan_budget` ; Claude Haiku 4.5 with the owner's key; photo not stored) → Claude answers JSON only → plausibility check → the photo and the numbers on screen → the person confirms → `POST /v1/bp/photo` (3 numbers only); the photo is deleted. A retake reason or "numbers wrong" is counted in `event_log` (`bp_photo_*` , code only). Code: `AiScan.kt` , `MonitorScan.kt` .

Neither way judges the values. Readings feed the charts, reports and the doctor's link; lab results feed the table in the Web Dashboard.

Application flows and permissions

Every flow of the application, step by step, and who is allowed to do what.

Application flows

#	Flow	Who	Steps	Data
1	Install, sign in	user	/download → install → Sign in with Google (OAuth, ID token checked by the Worker) → phone key registered	persons (fingerprint only)
2	Accept the terms	user	shown at first install, after every update and when the text changes → accept or the app closes	acceptances
3	Record by voice	user	Record (voice) or Scan (photo read on the phone) → check → Save → /v1/bp/voice or /photo	measurements, event_log
4	No AI	system	the Anthropic Scan was removed (terms v19): its calls answer 410 <code>ai_removed</code> ; nightly purge empties the old keys, costs and readings	person_keys, ledger, scans (emptied)
5	Screen lock	user	app lock: fingerprint, face or the phone's PIN; without any screen lock the app opens and shows a notice that leads to the settings (the user's responsibility)	none
6	See the report	user	Go to your full reports (end of Blood pressure) → Web Dashboard in a browser, Blood pressure page only: 7 days, one mean per day, the eight tiles of the app, morning/evening balance, extremes	read only
7	Share the report — app	user	Blood pressure tab, bottom → Download blood pressure report (PDF) → saved in Downloads, not opened (Android 9 and older: share sheet). The file stays on the phone; the server is not involved	none on the server
8	Share the report — web	user	Go to your full reports → Web Dashboard → cookie consent → PDF download (same A4) or Send to the doctor : Email or WhatsApp only; the PDF and a read-only link (7 days) go together, with a ready message in the phone's language; withdrawable	web_sessions, web_shares
9	Doctor's view	anyone with the link	open /s/... → charts and readings of that period, PDF; nothing else	read only
10	Delete, sign out	user	Admin → delete a reading, all, or the account; Sign out ends web sessions too	deleted at once
11	Owner's Admin tab	owner	App → Admin icon (last, owner only) → web console: 4 numbers, version switch-off; Observability: vulnerabilities, EU/CH compliance, problems and fixes (import outcomes stay in the log, not on the page)	counts only
12	Import a lab report	user	Lab results → PDF or photo (own reports only, terms v18) → a progress bar: Processing → Scanning → Uploading → outcome, then it closes; the app keeps an upload history by day (saved reports from the server, uploads that saved nothing on the phone, outcome codes only); in the background, all or nothing; same file or same date and day never twice; web table: one column per date, ↑↓ orange outside the printed reference, delete a date	measurements, lab_files
13	Automatic	system	nightly: readings > 365 days, lab results never (only the user deletes them), expired links and sessions, errors > 90 days deleted; every error logged	purge, error_log

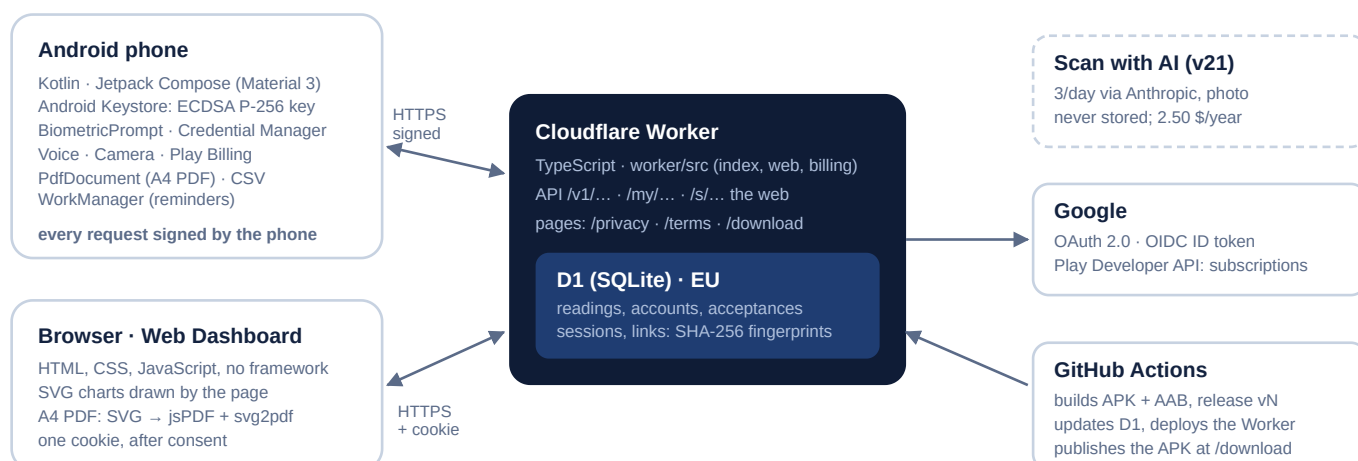
Who can do what

Action	User · app	User · web	Doctor · link	Owner	Developer
Record, delete own readings	✓	—	—	own only	—
See own charts and report, export PDF	✓	✓ (PDF)	the shared period	own only	—
Create or withdraw a doctor link	—	✓	—	own only	—
See another person's readings or report	—	—	—	never	never
See usage totals (never per account)	—	—	—	✓ Admin tab	via the owner's connector, for checks
Read the error log	—	—	—	✓ Admin tab	✓ to fix
Switch app versions off	—	—	—	✓ Admin, Actions	—
Change code, docs, deploy	—	—	—	decides	✓ through pull requests
See keys and secrets	—	—	—	✓ owns them	never

Enforced by the server: signed app calls, web session cookie, owner account, read-only expiring doctor links.

Architecture and technology

The components, how they talk, where the data live and how a new version is released.



Components

Part	Technology	Folder
Android app	Kotlin, Compose, minSdk 26, target 34	android/
Server	Cloudflare Workers, TypeScript	worker/src/
Database	Cloudflare D1 (SQLite), EU jurisdiction	worker/schema.sql
Web dashboard	HTML/JS, own SVG charts, jsPDF + svg2pdf	worker/public/my/
Sign-in	Sign in with Google (OAuth 2.0 / OpenID Connect) + phone key	Google.kt
Subscription	Google Play Billing + Play Developer API	Billing.kt , billing.ts
Build, release	GitHub Actions, Gradle, Wrangler	.github/workflows/
Guides, documents	HTML → PDF (Playwright)	docs/

Database tables

persons · accounts (pseudonymous account code, Google fingerprint, no email; subscription sub_*) · measurements pressure + lab results · lab_files keyed file fingerprints (no duplicates) · scans , ledger , person_keys left from the removed AI Scan, emptied every night · acceptances accepted terms (versioned evidence) · web_* web access and doctor links · settings , invites · error_log , event_log (import outcomes, codes only) · seen_sigs , rate_limits protection counters; security_snapshot_findings completed scan snapshots.

Security and data

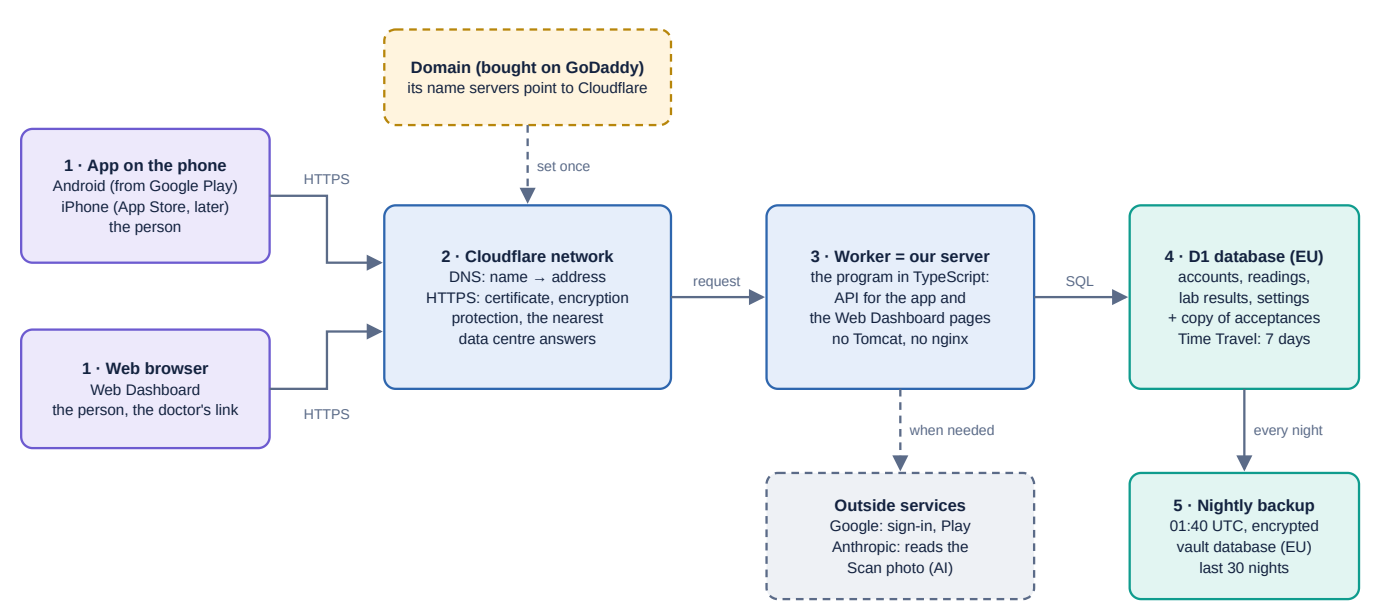
- Phone:** a private key that never leaves the phone signs every request; bad signatures, clocks off by over 5 min and replayed canonical messages are refused; mandatory Google nonce, single-use token and browser revocation on recovery.
- Limits:** hourly/daily caps on sign-ups, sign-ins, links, logs, voice (rate_limits , IP only as SHA-256). CSP and anti-framing headers; robots.txt .
- Web:** a 60 s atomically redeemed code in the # part of the address becomes the only cookie, hint_s (technical, 7 days, HttpOnly, Secure, SameSite=Strict), set only after consent. The app uses no cookies.
- Doctor links:** read-only, fixed period, expire after 7 days, withdrawable; no sign-in or cookie.
- One version only:** the app version travels with every request; the server accepts only the newest build (APP_VERSION); the app offers Google Play's in-app update. settings (app_min_version , app_blocked , app_off) stay for emergencies (Actions).
- Owner's area** (web, owner only): four numbers only (open vulnerabilities in the app code and in the Android/iOS libraries, open defects, compliance not fully covered), the way to Observability and the version switch-off. Never values or names.
- Subscription** (ready, off until Google Play): Play Billing, checked with the Play Developer API.
- Retention:** readings older than 365 days deleted every night; acceptances in an acceptance register.
- Personal data:** no name, no email (Google account only as HMAC-SHA256); codes, cookies, links only as SHA-256; no third-party API keys; documents read on-device; only lab results and a keyed file fingerprint sync. Source files never uploaded. Cloudflare D1 is EU-bound.

Releasing a version

- Change on a branch → pull request → merge into main .
- GitHub Actions updates D1, deploys the Worker, builds the APK and the AAB, stores APK/AAB in GitHub release v0.1.N ; /download redirects there.
- The site changes at once; the app updates by installing the new APK.

Runtime flow in five blocks

Who does what when the person uses HINT 365: phone or browser, Cloudflare, the Worker, the D1 database, the nightly backup.



Block	What it is	Who runs it
1 · App, browser	Android app (Google Play), iPhone later; Web Dashboard in any browser; every call is HTTPS, app calls signed with the phone's key	the person
2 · Cloudflare network	DNS (domain bought on GoDaddy, name servers set to Cloudflare), HTTPS certificate renewed automatically, DDoS protection, nearest data centre	Cloudflare
3 · Worker	our server, serverless: TypeScript run by Cloudflare's <i>workerd</i> engine (V8 isolates) for each request; replaces nginx (static files: Web Dashboard, APK) and Tomcat (the API); 2 cron triggers	code: us · engine and machines: Cloudflare
4 · D1 (EU)	SQLite-based managed database, jurisdiction EU; main data + copy of acceptances; Time Travel to any minute of 7 days (Workers Free)	Cloudflare
5 · Nightly backup	01:40 UTC, GitHub Actions: export of both databases, AES-256 with the owner's phrase, kept in the vault D1 (EU), 30 nights; restore and monthly drill by <i>Disaster recovery</i>	GitHub Actions

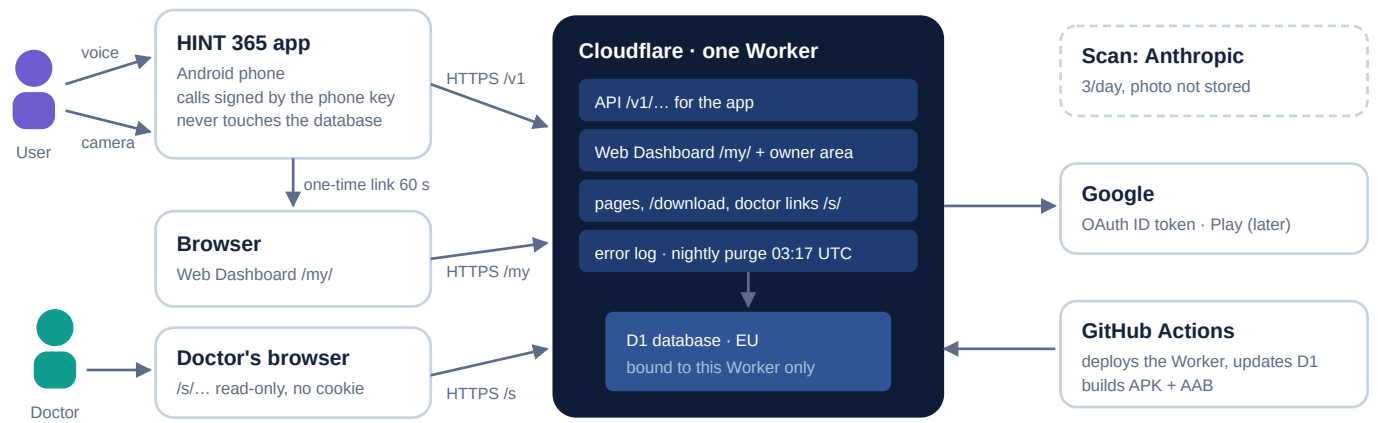
Hosting and runtime flow

Where every part runs, who talks to whom at run time, and how errors met by users are recorded.

Where it runs

Part	Runs on	Address	Operated by
Android app	The user's phone (APK; Google Play later)	installed from /download	the user
Web Dashboard (Blood pressure, Lab results; owner's area from the app)	Cloudflare Workers (static assets of the Worker) — not Vercel	personalhealthy-api.hint365.workers.dev/my/	Cloudflare, owner's account
API for the app, public pages, doctor links	Cloudflare Workers, the same single Worker	.../v1/... · /privacy · /terms · /s/...	Cloudflare, owner's account
Database	Cloudflare D1 (SQLite), EU jurisdiction	no address: bound to the Worker only	Cloudflare, owner's account
Source, CI/CD, releases	GitHub: repository, Actions, Releases	comevava76-web/personalhealthy (private)	GitHub, owner's account

Runtime flow only the Worker touches the database



Error log continuous testing and improvement

- **Server:** failed signed calls (400–599, except 402, 404, 426) and every exception.
- **App:** crashes and failed PDF files; at most 20 a session.
- **Web:** script errors, at most 5 a page; never on a doctor's link.
- **Short:** one row per day, source, code, place, version, with a counter; no free-text messages, so no reading value; 90 days.

Security docs/security/

- **Each PR:** 168 security and functional checks; the app must compile.
- **Nightly:** live probe; every library (full tree) on OSV.dev; Semgrep; gitleaks → Admin → Security, with **Fix** (Claude Code).
- **Deadlines:** critical 48 h, high 7 days, moderate 30 days. Dependabot PRs on Mondays.
- **Cost:** 0 \$ extra (GitHub free plan, owner's Claude plan, OSV.dev public).

Who uses it

- **Owner:** Admin tab → total errors (90 days); the details reach the developer through the daily issue.
- **Developer:** every morning a scheduled routine reads the report of the log and fixes (page 8); in working sessions it is also read after every release.
- **Table:** `error_log` in D1, written by `worker/src/errors.ts`; app side `ErrorReport` in `Core.kt`; web side in `app.js`.

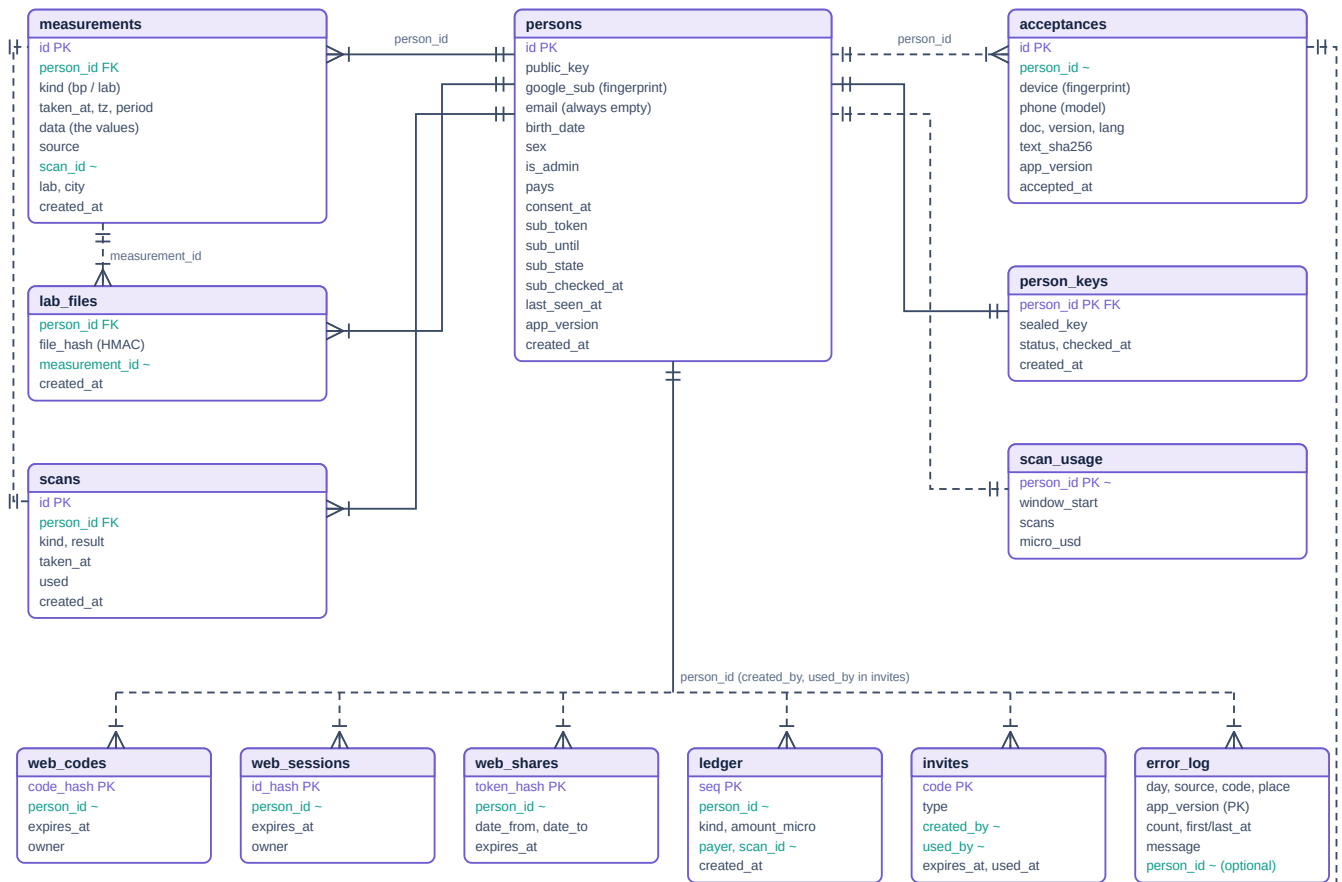
Data protection GDPR · Swiss FADP

- **Health data,** pseudonymous (not anonymous): the GDPR and the Swiss FADP apply. Basis: explicit consent (Art. 9(2)(a)) at sign-in and in the terms (v15, section “Data protection”).
- **Roles:** owner = controller; Cloudflare = processor; Google and Anthropic (Scan photo, processor) in the US under their transfer safeguards; the photo goes only to the owner's provider (terms v21).
- **Rights:** access, correction, deletion (in the app), a copy, restriction, objection; answer within one month.
- Gaps still open and the record of processing: `docs/compliance/gdpr.md`.

Database: tables and relations

Cloudflare D1 in the EU: **personalhealthy**, plus **personalhealthy-backup** for the copy of the acceptances. Only the Worker reads and writes them.

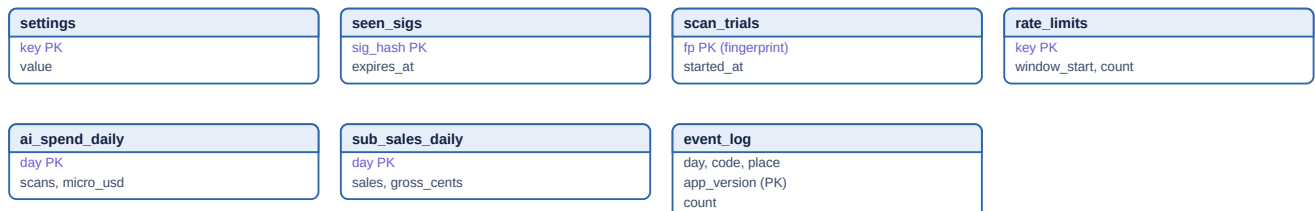
Personal data: every table linked to persons (one account)



Backup database · Security (Observability)



Tables without relations

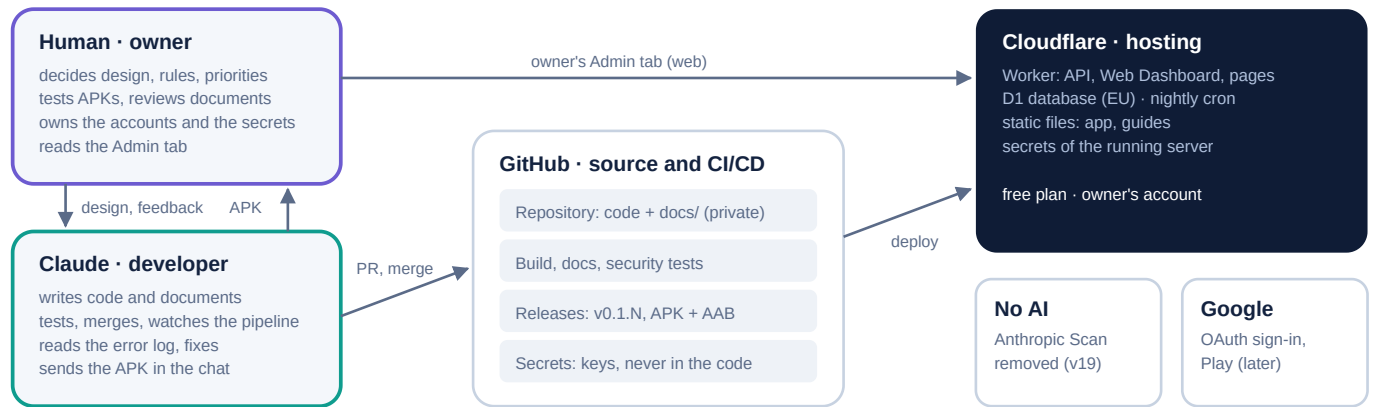


PK key of the row · **FK** foreign key declared in the database (solid line, deleted with the account) · ~ same id kept by the Worker, no constraint (dashed line). Line ends: two bars = exactly one, crow's foot = many. Deleting an account removes every personal row except the acceptances (proof). Contents and retention of each table: general document, section 4.

Roles and delivery

Who does what — the architect, the developer, the repository and the hosting — and how a change goes from the chat to the phone.

Roles people and platforms



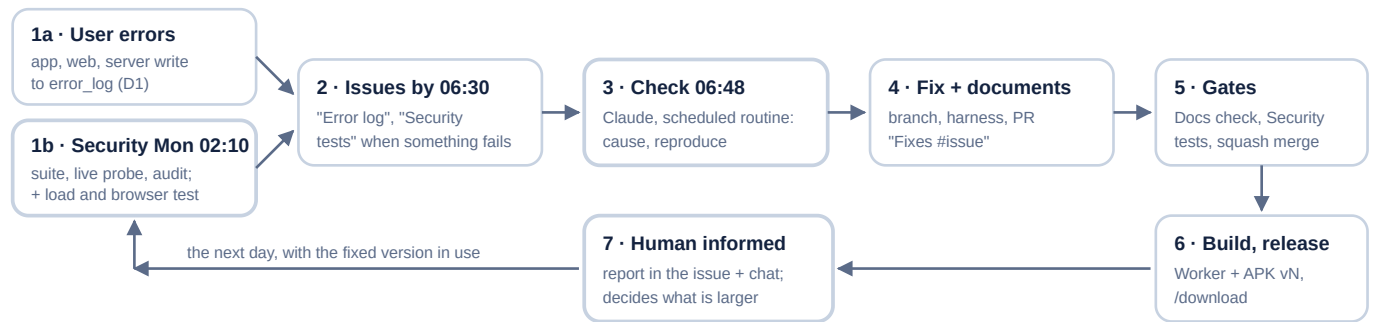
Responsibilities

Activity	Human · architect	Claude · developer	GitHub	Cloudflare
Design, rules, priorities	decides	proposes, asks when unclear	—	—
Code and documents (docs/)	reads	writes, same commit	stores	—
Tests before merge	—	runs	Docs check, Security tests	—
Merge, build, release	delegated	merges, watches	builds, releases	—
Hosting, database, cron	owns the account	—	deploys	runs
Keys and secrets	owns	never sees them	stores	stores
Trying the app on a phone	tries every APK	sends it	—	serves /download
Errors met by users	sees them in Admin	analyses, fixes	—	stores the log

Delivery, from the chat to the phone

- Decide** — Human describes the change in the chat; unclear points are asked, not guessed.
- Build and test** — Claude on `claude/...`: code and documents in one commit; harness, pages in Chromium, documents rendered.
- Merge** — pull request, *Docs check* and *Security tests* green, squash merge into `main`.
- Release** — local tests → privately signed APK/AAB → Worker + same-revision APK promotion → release record. No missing-key fallback. D1 tables, Worker, APK/AAB `v0.1.N`, `/download`; Claude checks it and sends the APK.
- Try** — Human tries the APK; feedback starts the next round. Rollback: revert the pull request.

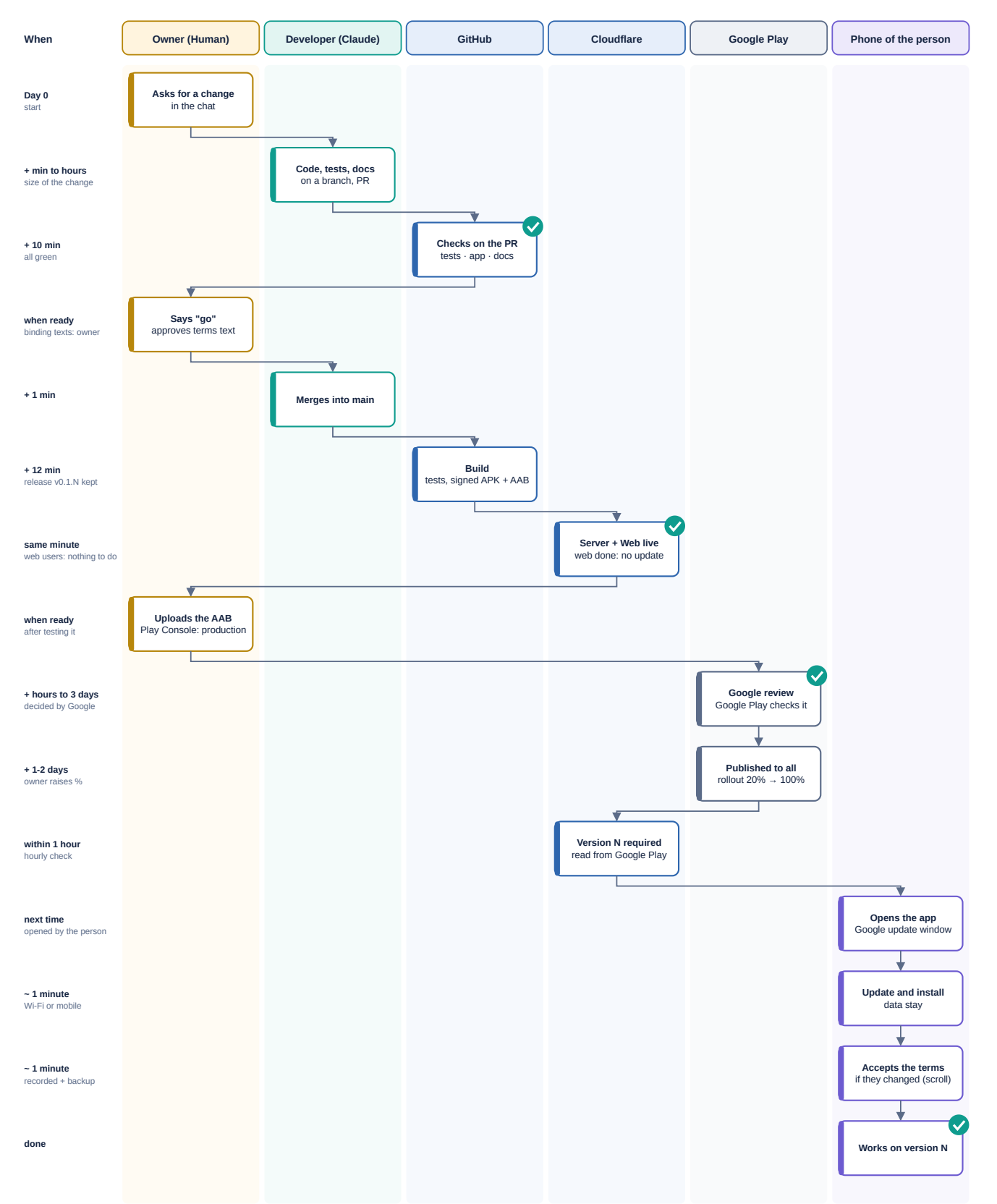
Continuous improvement loop proactive: every day, without being asked



Security tests, vulnerability scan, Dependabot and who pays for the loop: page 6. Larger fixes wait for the human.

Release and deployment - Android

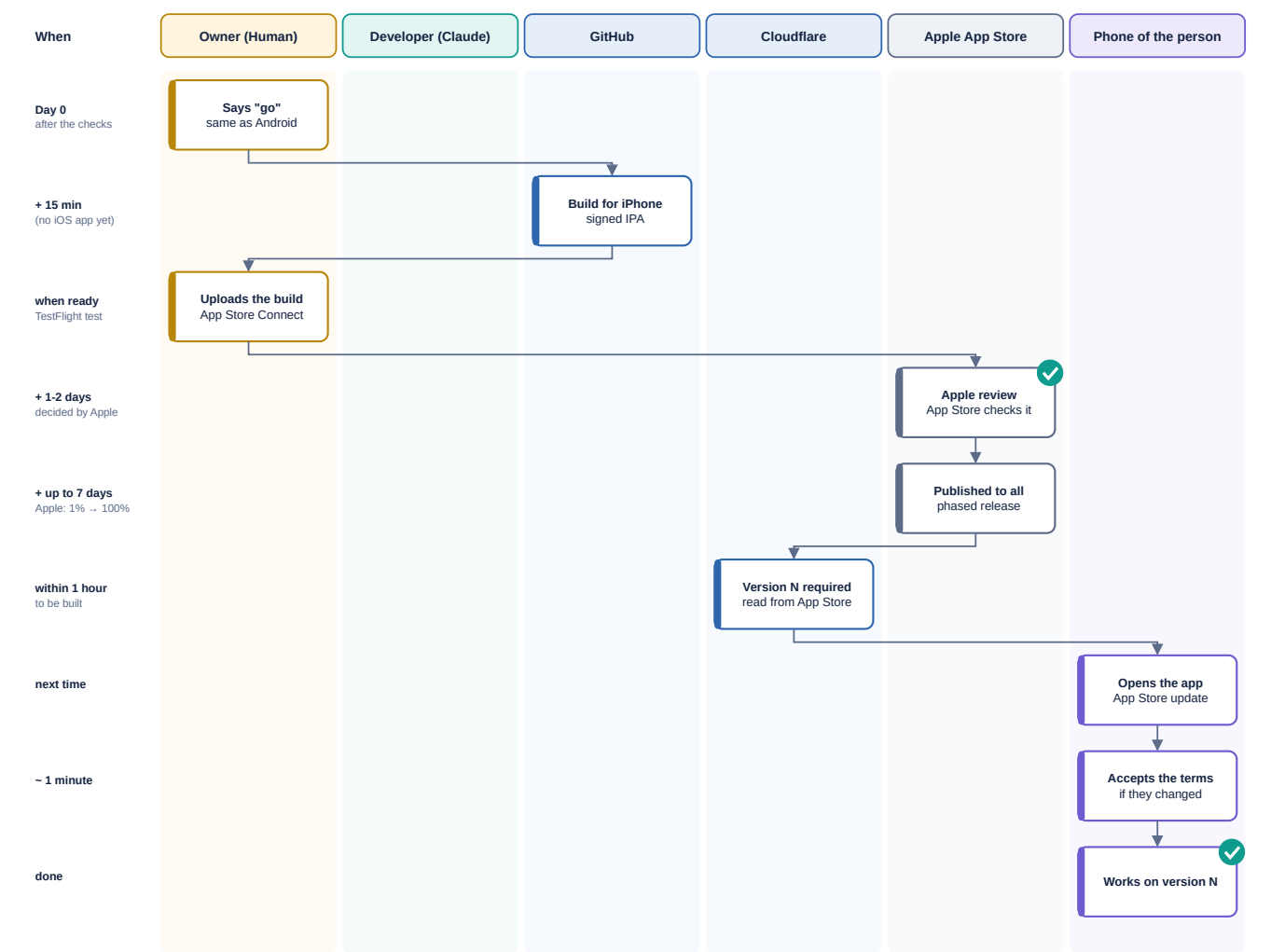
Who does what, and when, from the owner's request to the person working on the new version. Server and Web Dashboard are live minutes after the merge; the app waits for Google's review.



✓ = check passed. The Worker reads every hour the newest version Google Play has published to everyone (production track, status completed) and only then requires it (`app_live_version`); before Google Play, the build just deployed. Phones with automatic updates get it from Google Play without opening the app.

Release and deployment - iPhone and roles

The same release through the Apple App Store, when the iOS app exists (today there is none), and the part each actor plays.



Who does what

Actor	In a release	Time
Owner	asks, says "go", approves terms changes, uploads AAB / iOS build, raises the rollout	when ready
Developer	code, tests, docs, pull request, merge, deploy check, link in the chat	minutes to hours
GitHub	checks on every PR; Build: signed APK + AAB, release v0.1.N	~10 + 12 min
Cloudflare	server and Web Dashboard live; hourly: version live on the store becomes the only one accepted	same minute; ≤ 1 h
Google · Apple	review ✓, then publish to a share of people, then to all	hours–3 days · 1–2 days; Apple up to 7 days
Person	opens the app, updates in the store's window, accepts the new terms (scroll to the end), goes on	~2 min